Circular Motion: Lab Analysis

SYNOPSIS

Note 2 said that the centripetal force is whatever real force happens to be pointing at the centre, and that $F_C = mv_t^2/r$. This lab tests the second half of that: **how do we show that F_C is proportional to v^2?**

The apparatus is a rubber stopper on a string, swung in a horizontal circle above your head, with the string running down through a short tube and a hanging mass on the lower end. The trick is the tube. **The tension is the same everywhere along one string**, so whatever the hanging mass weighs, that is the tension pulling inward on the stopper — and the hanging mass is something you can simply choose, by counting out washers.

So you set the centripetal force, you measure the speed it produces, and you see whether doubling the force multiplies the speed by √2. That is the whole experiment.

BEFORE YOU READ ON — PREDICT

You are swinging the stopper steadily and someone quietly adds washers to the hanging mass, doubling it. To keep the stopper going round the *same* circle, do you have to swing it faster, slower, or the same? Write one sentence, and if you can, a factor.

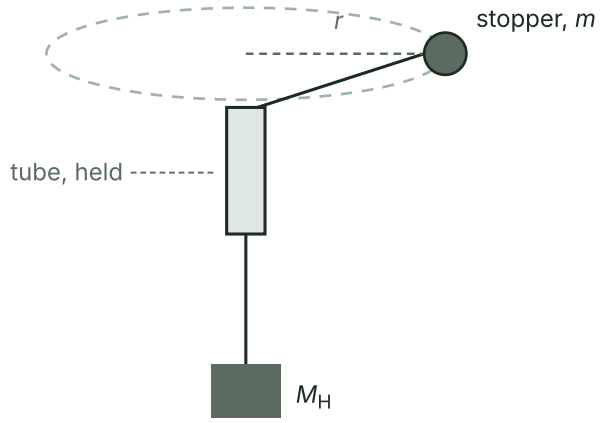
PERFORMANCE EXPECTATIONS

Draw the side-view and top-down free-body diagrams for a conical-pendulum apparatus and identify the centripetal force. Use the fact that the tension in an ideal string is the same throughout to set F_C from a hanging weight. Measure a tangential speed by timing revolutions. Design a test of $F_C \propto v^2$ and say what graph would show it.

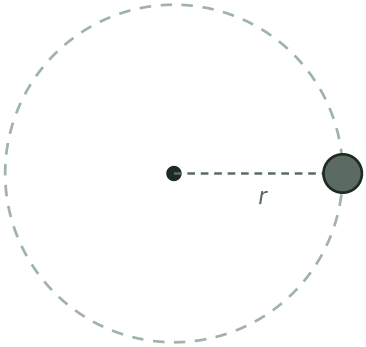
The apparatus, and the one idea that runs it

Two free-body diagrams again — one for the stopper and one for the hanging mass — and one observation that connects them and makes the whole thing work.

side view



top-down view of the stopper



Draw both free-body diagrams. On the side view, every force on the hanging mass and the tension in each part of the string. On the top-down view, the horizontal forces on the stopper and its velocity. Then write, on the line underneath, the one sentence about the string that ties the two diagrams together.

1. Draw the free-body diagram for the hanging mass and write Newton's second law for it, given that it hangs *still*. What does that make T?
2. Now the stopper, seen from above. Which force on it points at the centre of the circle, and what is its size? Put those two results together into one statement about F_C.
3. Why does it matter that the string runs through a *tube* rather than being tied to something? What would change if you tied the lower end to the bench instead?

How do we show that F_C is proportional to v^2?

You can set the force and you can measure the speed. The question is what to do with the pairs of numbers so that the answer is visible rather than argued.

4. You cannot time one revolution accurately by hand. Say how to measure v_t well, given a stopwatch, and write the formula you would use.
5. You vary F_c by changing the number of washers, keeping r fixed. Starting from $F_c = mv_t^2/r$, what should you plot against what to get a straight line, and what is its slope?
6. Suppose m = 25 g, r = 0.80 m, and you hang four washers totalling 0.100 kg. (A) What is F_c? (B) What speed should the stopper be going? (C) How long should 20 revolutions take?
7. You then double the hanging mass to 0.200 kg at the same radius. By what factor should v_t change, and by what factor should the time for 20 revolutions change? Check against the numbers.

Looking ahead

The same apparatus answers a second question for free: hold the washers fixed and change r instead, and $F_c = mv_t^2/r$ says the speed should go as $\sqrt{r}$. And the honest caveat is worth saying out loud — the string is never quite horizontal, because something has to hold the stopper up against gravity, so the real apparatus is a **conical pendulum** and the tension is very slightly larger than $M_H g$. Note 2 has the conical pendulum worked in full.

BACK TO YOUR PREDICTION

Faster, by a factor of $\sqrt{2}$. Doubling the hanging mass doubles the tension, which doubles the centripetal force; and since F_c goes as v^2, doubling it multiplies the speed by $\sqrt{2} \approx 1.41$, not by 2.

If you said “twice as fast”, that is exactly the assumption this lab is built to test, and the stopwatch will disagree with you by about 5 seconds in 18.

Check yourself

1. A hanging mass of 0.100 kg sets the tension in the string. What centripetal force does that put on the stopper, in N?

_____ N

2. The reason the hanging weight tells you the centripetal force on the stopper is that

- ○ the stopper and the hanging mass have the same mass
- ○ the tension is the same everywhere along one string
- ○ the tube pushes on the string with an equal force
- ○ gravity on the stopper equals gravity on the hanging mass

3. To test $F_c \propto v^2$ with a straight-line graph you should plot

- ○ F_c against v_t
- ○ F_c against v_t^2
- ○ v_t against F_c^2
- ○ F_c^2 against v_t

4. You double the hanging mass at the same radius. By what factor does the stopper's speed change?

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Your notes

BEFORE THE LAB

Guided notes — blank (PDF)

print or open on a tablet; record your prediction, then work the items as the video goes through them.

AFTER THE MODULE

Completed notes (PDF)

released when Module 5 closes.

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